

RAG Pipeline Evaluation and Optimization Handbook

1. Evaluation Methodology

Evaluating a Retrieval-Augmented Generation (RAG) system requires assessing two key aspects: Retrieval Quality and Generation Faithfulness. Retrieval quality measures whether the vector store returns chunks containing the ground-truth answer. Generation faithfulness measures whether the LLM answers strictly based on the retrieved context without hallucinating external information.

2. Groundedness and Citation Testing

Groundedness ensures that every claim made in the generated answer directly maps to a retrieved source chunk. Auditable source citations (e.g., referencing source filenames like 'ai_rag_overview.txt') are verified during test suite execution. Failure cases typically occur when retrieval top_k is set too small or when chunk boundaries split key entities across separate fragments.

3. Mitigation Strategies

To mitigate context fragmentation, implement fixed-size chunking with overlapping windows (e.g., 800 characters with 150 characters overlap). To mitigate hallucination, enforce strict prompt system instructions requiring the model to state 'I don't know' if the answer is not present in the context.